

# VRtoER: When Virtual Reality leads to Accidents

A Community on Reddit as Lens to Insights about VR Safety

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Figure 1: Compared to laboratory settings (left), the use of VR in domestic settings poses many dangers.

## ABSTRACT

Virtual Reality (VR) has gained a relevant market share as a product for consumers in the gaming industry. Compared to other hardware, VR headsets isolate users from reality visually to create immersive experiences. However, when not used correctly, VR can lead to accidents, for instance, when colliding with objects in the real environment. A community on Reddit has formed around the topic of VR-related accidents and injuries called r/VRtoER, in which users upload media of accidents and injuries through the use of VR. 82 posts of the community were analyzed to understand 1) the cause for the accident, 2) why it happened, and 3) how it could be prevented. Preliminary findings are presented based on the analysis. The results show that in 26.83% of analyzed cases, the environment was not suitable for VR use. The analysis has also shown six cases in which VR use led to child endangerment.

## CCS CONCEPTS

• **General and reference** → Surveys and overviews; • **Human-centered computing** → Virtual reality.

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## KEYWORDS

Virtual Reality, Safety, Reddit, Accidents, r/VRtoER

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## 1 INTRODUCTION

With the emergence of cheaper and commercially available Virtual Reality (VR) hardware since the 2010s, VR has made the move into the consumer sector and is increasing its popularity. VR has become an important market for the gaming industry, and thus, is used outside of research labs in users' homes. With a growing market for VR, new games and devices are being introduced to the market steadily. Through the hype around the concept of "Metaverse", VR has gained even more attention, as it is seen as a key way to enter and experience the metaverse due to its immersive features [6]. By isolating the visual (real) world through VR head-mounted displays (HMDs), VR lets users experience Virtual Environments (VEs) in a highly interactive and immersive way. However, when VR systems are used in home environments, there is a risk that the interactive nature of VR will clash with the existing domestic facilities (Fig. 1). There is an increasing number of accidents through the use of VR, with insurance companies also indicating a rise in insurance claims from VR users [1].

This emergence of VR-related accidents and injuries is a relevant challenge for Human-Computer Interaction (HCI) as the technology isolates users visually (and maybe auditorily) from his/her contextual environment during the use of VR. VR-related injuries can have a life-changing traumatic impact on the users' life [5]. However, from a research perspective, it is challenging to find out how the accidents happen and what the main causes are, as they occur in the user's homes. In [2], Dao et al. have presented an analysis of 233 VR fail videos retrieved through *youtube* to gain insights about the topic. For their analysis, they have categorized the fails, tried to understand the causes of the fails, and derived design implications for VR systems. In [11], Tseng et al. have explored the risks and potential harm of Virtual-Physical Perceptual Manipulations (VPPMs) that alter the human perception of own physical interactions in Extended Reality (XR) settings. In their analysis, they provide a classification of potential attacks on XR users and characterize physical harm through these attacks.

With the approach used here, a contribution to the research of VR fails and harmful VR use is made by concentrating on VR-related accidents and VR-induced injuries. On the online platform Reddit, the community ('subreddit') *r/VRtoER*<sup>1</sup> has formed for the purpose of sharing videos, photos, and stories about VR accidents and VR-induced injuries ("VRtoER" as the acronym for "[From] Virtual Reality to [the] Emergency Room"). The name of the community is not meant literally from VR to ER but as an exaggeration for VR-related accidents. In the description of the community, it refers to sharing "VR injury only". The sub was created as a follow-up after a user replied solely "*r/VRtoER*" on another comment in the subreddit *r/gaming*: "*Will be exchanging that VR headset for an ER neck brace*" on a VR fail video in which a woman runs into a brick wall/chimney while using a VR headset<sup>2</sup>. Among others, the community consists of VR enthusiasts, VR experts, VR users, and enjoyers of fail videos. In the subreddit, users share their own injuries and accidents and those of others in form of text contributions, photographs of injuries, and videos. The content of *r/VRtoER* can mostly be defined as "accidents or injuries that resulted through or with the use of VR headsets". With the content in *r/VRtoER*, the community might contribute to understanding accident patterns in the domestic use of VR systems. For this, 200 submissions of the subreddit have been analyzed to understand 1) the incidents of accidents, by trying to understand what happened and how it happened, 2) the cause of accidents (why the accident happened), by trying to understand the reason for the incident, and 3) how such accident could have been prevented. The research contribution is twofold: on the one hand, the analysis of the data shows how social media data can help to understand needed safety aspects of VR technology, and on the other hand, the results are presented based on the analysis of the community's content. The purpose of this paper is to show researchers, game designers, and users the dangers of playing VR content and what could have been done to avoid these dangers.

<sup>1</sup>[www.reddit.com/r/VRtoER/](http://www.reddit.com/r/VRtoER/)

<sup>2</sup>Comments on Reddit that consist solely of a name of an existing or non-existing subreddit are supposed to show other users under which subreddit the submission would fit well.

## 2 VR INTERACTION RELATED SAFETY ASPECTS

Commercial VR systems are delivered with several consumer warnings and safety features. For instance, the company *Meta* recommends users to use the stand-alone VR headset Quest 2 only indoors, to make enough room for the VR interaction space, and has set up a 'safety-center'-website with multiple safety instructions when using the headset. However, domestic environments may differ in space and are individually furnished. As a result, the use of VR systems is always subject to the compromise of safety aspects to be followed and the limited space in the domestic facilities. Room-Scale VR allows moving around physically in the VR space. To prevent collisions with objects, consumer VR systems such as the HTC Vive, Valve Index or Meta Quest 2 provide a safety feature to define the VR play area beforehand called *Chaperone* (HTC Vive, Valve Index, etc.) or *Guardian* (Quest). In principle, this safety concept consists of visualizing the boundaries of the play space in VR when the player is approaching or even leaving those boundaries. For instance, the Quest 2 switches to the *see-through* mode of the headset, allowing players to see their real environment (Fig. 2). However, there is the passersby problem, in which people, animals, or objects can enter the VR tracking space (play area) without the player noticing [12]. Such tracking space invasion can result in players hitting and hurting other humans passing by. Other safety features are wrist straps for the VR controllers to avoid controllers from flying through the room like projectiles when losing them during fast movements or the use of wireless or stand-alone systems to eliminate the risk of tripping over cables.



**Figure 2: Concept of *Guardian*:** As soon as the player goes beyond the boundaries of the guardian, the *see-through* mode of the headset starts and the player sees the real environment

## 3 METHOD

To generate the data set, a script using the *Python Reddit API Wrapper* (PRAW)<sup>3</sup> as well as *RedDownloader*<sup>4</sup> packages to scrape the data of *r/VRtoER* was implemented. Then, the media and the comments from 200 of the most upvoted posts (or 'submissions') of all time in the subreddit were downloaded. For each post, a number of different parameters were extracted, the most relevant beyond media files and comments were submission text, title, score (upvotes), number of comments, and date of submission.

For the qualitative analysis, the data set was first scanned for duplicates, deleted submission files that could not be retrieved otherwise, and meta-content like memes about the community or

<sup>3</sup>PRAW: The Python Reddit API Wrapper, <https://praw.readthedocs.io/en/stable/>

<sup>4</sup>RedDownloader 4.2.0, <https://github.com/JackhammerYT/RedDownloader>

ads. In total 42 data entries were removed (21%) in this first filtering step. Then, in a second step, the media files were scanned to gather further context information like the general happening (for videos) or if further information was provided (for photographs and videos). Submissions showing images of injuries or hit objects were removed when no further or too little information about the incident was provided. For instance, image submissions showing bloody fingers or punched computer monitors were excluded from the analysis when no further explanations of the incident were given by the submission authors. As this was the case for the majority of image submissions, due to this exclusion criteria the document set was reduced to 94 data entries (47% of the original data set). During this filtering step, further information about the whole happening was documented for each data entry. If the contextual information of the submissions allowed it, additional information was documented like which object was hit, the used game/app, the used VR device, the locomotion technique, what was recorded (for videos), or if spectators were present. In addition, first inductive categories were created describing the accident incidents.

Then, in a third scan, the videos were also rated for their plausibility and if they were 'fake' or not. For the fake assessments, the comments beneath the videos were scanned, especially when the videos showed somehow suspicious incidents (e.g., misuse of hardware). In total, twelve data entries were removed. Five videos were rated as fake or staged, and for seven videos a rating was not possible due to missing contextual information. Thus, with this strict filtering process, the final data set was reduced to 82 entries (41% of the original data set) and contained submissions between May 2019 and October 2022.

The data set was then further checked for contextual information. Repetitive themes and categories were noted. Specifically, it was documented (1) what happened during the incident, (2) which actions led to it (how the incident happened), and (3) what was the cause for the incident, i.e. why the accident happened. Based on this qualitative process, more general categories were defined for the incidents as well as for the accident causes.

## 4 RESULTS

To analyze the submissions, the sequences of events in the videos were reconstructed, as the videos allowed to interpret a cascade of events. This was relatively easy for most of the submissions. However, there were some videos in which the camera panned slightly during a relevant moment, making it impossible to clearly understand the accident sequence. For instance, in one video, a player walks a few steps and then falls forward (T35). The camera work makes it unclear whether the player stumbles over something or simply loses her balance in this incident. In other examples, the videos showed only the VR screen capture. Although these allow roughly to understand what happened, the view of the real world was missing. In some cases, the incident could be made more comprehensible by the players' statements or by additional information posted in the players' comments. For example, in a VR screen capture of a user playing *RecRoom*, it was possible to tell from the player's reaction that it was a repetitive action: "[Sound of an object being hit] Ow, Ow, Ow, oh my god. what did I just do again? Oh, I'm bleeding, [small pause] again?", and that the player hit a light

**Table 1: Objects VR Users Collided with**

| Category                      | Incidents | Percentage |
|-------------------------------|-----------|------------|
| Furniture                     | 30        | 36.59%     |
| Floor                         | 19        | 23.17%     |
| Room Boundaries (e.g., Walls) | 16        | 19.51%     |
| Ceiling (e.g., lamp, fan)     | 5         | 6.10%      |
| Human                         | 5         | 6.10%      |
| Unknown                       | 3         | 3.66%      |
| PC                            | 2         | 2.44%      |
| None                          | 2         | 2.44%      |

bulb (T99). From the comments provided by the submission author, the cause of the accident was rated as "Room space not suitable" for VR use, as the player hit the nearby ceiling lamp during play. With this approach, the analysis made it possible to understand the cause of action that led to the accident incident and how the accident could have been prevented. For instance, if a video showed a person using VR in a space that is not suitable for VR use, i.e. because it is too narrow or too near to walls, but the accident happened because the player was being pushed by a spectator, then the prevention mechanism would not be "use a suitable space" but "do not push players in VR". Depending on the video and provided data it was also possible to extract which VR headset was used, how many spectators were present, or which game was played.

### 4.1 Descriptive Data

In total, 82 videos and photographs were rated as non-fake and gave enough contextual information to analyze the incident. Regarding the incident categories, in 25 cases (30.48%) the player fell over, whereas in six of these cases (7.31% of total) the player fell *head first* into a wall or another object. Other categories were running into objects or a wall (21.96%), hitting the head on furniture (9.76%), hitting another human being (9.76%), hitting objects above or in front (14.64%), jumping into objects or a wall (4.86%), or other cases.

Regarding the different objects that the player collided with or that the player hit with the controllers, the objects that had the most impact on the accident were singled out during the analysis (Table 1). In most cases, the players hit some sort of furniture (e.g., chairs, tables, TV), fell on the floor, or collided with room boundaries (e.g., walls, column pillars). In two cases no object was hit. In both of these cases, the VR space was invaded. In the first case, a *near miss* could be observed as a child was almost hit by the VR player, and in the other case, a cat bit the VR player in the foot.

In trying to understand the root cause of the accident, six main categories were identified (Table 2). Most accidents could be understood as due to immersion, referring to situations in which players got scared and made excessive movements or situations in which they seemed to have forgotten their presence in their real environment. In 22 cases the room space was not suitable for the VR experience, in 16 cases sensory-motor problems (e.g., losing balance) seemed to have caused the accident, followed by an invasion of VR space, spectator engagement, or hardware not properly used. In one case the cause could not be understood. In the following subsections the categories *Immersion*, *Room Space*, *Invasion of VR*

**Table 2: Main Categories for Accident Causes**

| Category                           | Incidents | Percentage |
|------------------------------------|-----------|------------|
| Immersion                          | 27        | 32.93%     |
| Room space not suitable            | 22        | 26.83%     |
| Sensory-motor problem              | 16        | 19.51%     |
| Invasion of VR space               | 8         | 9.76%      |
| Spectator engagement               | 5         | 6.10%      |
| Hardware not properly used / setup | 3         | 3.66%      |
| Unknown                            | 1         | 1.22%      |

*Space*, and *Spectator Engagement* will be discussed. In addition, the cases of child endangerment through VR use will be explained, before discussing the content of the subreddit from the community's users' point of view.

## 4.2 Immersion

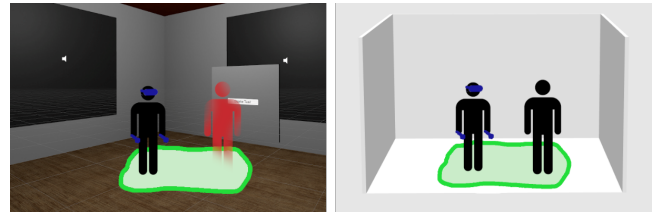
This category was used as an umbrella term for incidents in which players seemed to have forgotten their presence in or the boundaries of the real room because they were immersed in the virtual world or had experiences of flow. For instance, in at least five cases the players got scared from what they experienced in VR (e.g., Zombie appearance, fast oncoming bus) and had a flight reaction. In all these cases the players started running into the wall or other objects. In at least two cases the players tried to lean on a virtual table, causing them to lose their balance and fall forward on the ground. However, a total of 12 accidents involved jumps during a VR experience called "*Richie's Plank Experience*". In this VR experience players are walking on a plank at the edge of the top of a skyscraper. In many of these cases, this VR experience has led players to decide to "jump down". But instead of flying harmlessly towards the ground in the virtual world, as players seem to expect in the moment, players make a physical jump in their real environment, causing them to hit their head on the floor, jump into the TV, a Christmas tree, or other objects. These videos repeatedly lead to discussions in the subreddit, questioning the decisions and actions of players, or accusing these videos of being fake, as one user described: "*I cannot comprehend how this man just decided to L E A P*" (T1).

## 4.3 Room Space

The second category refers to room spaces that are not suitable for VR interaction. In five cases players hit an object on the ceiling (e.g., a lamp, fan, light bulb). Even if such safety systems as *Guardian* are used, they usually have their anchor points on the ground in order to prevent tripping over objects or colliding with objects. In six cases the players hit an object or a human in front of them because of a VR space that was too small. In another six cases, the players ran into objects like the TV or other furniture. These cases suggest that no *Guardian/Chaperone* was used, as these systems could prevent running into the TV by switching to the see-through mode as soon as the player hit the *Guardian* border. The case of running into and breaking the TV reoccurred so often in the community that a user commented on one of the videos: "*Sometimes I think the whole VR gaming industry is a front by the TV companies to drive more sales*" (T41).

## 4.4 Invasion of VR Space

In the data set 7 accidents and one near miss were identified as invasions of the VR space. In such cases, people (or animals) walk into the VR play area of the player and put themselves in danger, since the player, depending on the level of immersion and sensory impressions, does not notice the intrusion. In three of the cases analyzed, the person was hit in the head with the controller. In order to avoid such accidents, a physical barrier of the VR space could help, for example, to keep animals or children out of the VR space. Another method would be to integrate passersby spotting into the headsets [7, 12]. In fact, the Quest 2 headset has such a tracking method integrated (called 'Space Sense'), and users can see the contours of the invader. In five cases the player started hitting something fast (e.g. during a boxing game) and passersby spotting showing contours of the invader might not be effective to prevent such accidents.



**Figure 3: Spotting passersby, when someone enters the play space their contour is visualized in VR.**

## 4.5 Spectator Engagement

While there were spectators in most of the videos, five cases were identified in which the spectator engagement was probably the cause for an accident, in all cases during the 'plank experience'. In three of these cases, the VR players were pushed by one of the spectators, leading to heavy accidents. In one case, the player lost balance through the push and fell head-first into a wall. In the other two cases, spectator engagement probably led to a misunderstanding in their communication. In the first case, an older woman is doing the plank experience and the spectator repeats "*step to the right. you'll just float, you'll float*" which causes her to leap forward, against the wall (T24). In the second case, a young child asks the people around him "*can I just run and jump off?*". After receiving a "*yeah*" as an answer the child starts running straight into the TV, although the spectators still tried to stop him (T28). Spectators can have a vital role in the VR experience of users. In [9], O'Hagan et al. described interactions between VR users and bystanders in three categories: "*Coexisting, Demoing, and Interrupting*". In addition to the negative impacts of bystander interventions on VR users, bystanders can also warn VR users of potential dangers. However, due to the lack of sensation, such cases are probably less often captured on video and uploaded to communities such as r/VRtoER.

## 4.6 Child Endangerment

In the data set six quite shocking cases were found in which children were endangered through the use of VR, either by playing VR despite being too young or because they got hit by others. Only in



one of the cases, no accident occurred (near miss). In four cases, children were hit by accident by an adolescent or another child due to the invasion of the VR space. In the last case, a child jumped during the VR experience and fell down head-first. In two cases it can be seen that the children are clearly younger than 13 years and therefore it is to be questioned if they should use these devices at all. To avoid such cases, both the manufacturers of the devices and the parents are asked to intervene. On the producer side, even better tracking of VR space invasion could be used with stronger consequences, like a faster switch to pass-through mode, etc. Also, more education about VR use in homes with children seems necessary. Parents should ensure that the VR play area is not accessible to children to protect them from any harm.

#### 4.7 User Discussions about the Content and the Focus of r/VRtoER

A lot of content was followed by a discussion about whether the content submission had an existing right in the subreddit. These discussions serve to define the purpose and the common focus of the community more precisely. For instance, under the submission of a video in which a player kicks slightly an open drawer in front of him (T185), users disagreed with the submission, as it would not show “an accident”. However, the author of the submission clarified: *“This sub is for posting accidents that happen when a person is in VR like hitting stuff when you’re in VR. Not just cases where the person has to go to the hospital. And there was a small accident he if you didn’t notice. He hit stuff when if VR unaware of the surroundings in a very stupid way and that is what this sub is about.”*. The less dangerous accidents are also regularly picked up in the subreddit via memes or other meta-content. For instance, submission T59 displays an abstractly drawn meme in which a player plays in a 1.2 x 1.2-meter apartment, gets scared heavily, and hurts his hand, indicating satirically that (1) in many submissions the use of VR systems occurs in play spaces that are too small, (2) the impulsive and excessive reactions of many users while playing (e.g., getting scared and running in the opposite direction), and (3) also satirically referring to the huge number of submissions showing minor injuries of the hand.

## 5 LIMITATIONS

Although strict parameters were applied about which videos to include in the analysis and videos that were even slightly suspicious were removed, there is a residual probability that some of the analyzed videos might have shown faked content. To exclude fake videos, the plausibility of the video sequences was evaluated, and eventual explanations by the submitters of the content were included in the assessments. In addition, the comments under each suspected submission were evaluated. With the used approach, twelve data entries were excluded that showed misuse of VR hardware on purpose or where it remained unclear if the videos were faked or not. Another limitation lies in the videos themselves. Although videos provide a lens on the actual context of use and the happenings of VR-related accidents, they usually don’t show the aftermath of the accident or explain the injuries resulting from the incidents. Also, the data set was limited to the 200 most upvoted submissions of all time, thus probably adding a bias to the data set

towards more shocking, funny, or discussion-worthy submissions. In addition, using the most upvoted submissions, a high number of duplicates had to be removed from the data set. Lastly, another limitation is that the qualitative analysis was conducted by only one person. In assessing the content and meaning of data, qualitative analyses can benefit from the discussions of multiple researchers during the analysis process (e.g., [8]). However, when assessing the content of the videos, the comments of other users below the submission were also included whenever it seemed appropriate.

## 6 CONCLUSION AND FUTURE WORK

In this paper, preliminary results are presented of an analysis of data gained from the subreddit r/VRtoER regarding VR-related accidents and injuries. The results have shown how data from online communities such as r/VRtoER can be used to gain insights about such accidents, which happen in users’ homes and are not observable otherwise. In the final data set, 26.83% of the 82 cases were classified as being cases of “unsuitable room space” in domestic facilities for VR use, highlighting the challenge of safe VR interaction in home environments. The results have shown cases where the issue of VR safety is a matter of negotiation between the immersion possibilities and certain safety mechanisms (e.g., the use of Guardian). However, in certain categories, such as “No suitable Room Space” it might be difficult for users to find enough space in their homes for the safe use of VR experiences. In such cases, technology incorporating intervention designs like the use of Electrical Muscle Stimulation (EMS) might be a solution to prevent users from hitting their surroundings. In [4], such an EMS prototype has been evaluated while participants used VR. The results showed promising results to restrain the arm movements of users. The results will help researchers and practitioners to better understand VR-related accidents. The work also expands knowledge of existing research in this area [2]. Compared to the approach used in [2], the results presented here were gained from a community of VR users, who share their knowledge and views in the comments below the submissions. For future work, an analysis of the comments could provide even more detailed insights into the accident sequences. Whereas this paper focused on accidents through the use of VR, comparable challenges arise with Augmented Reality (AR), especially mobile AR applications and games. With AR, computer-generated objects are shown in the real world of users, as they are not isolated visually, but still can retrieve information from their peripheral vision. However, in mobile AR scenarios, dangers can result from environmental contexts like infrastructure. There are reported cases where players have gained injuries while playing *Pokémon GO*. For instance, one player fell onto a railway while playing the game [10]. In another example researchers have linked playing *Pokémon GO* while driving with increased car accident numbers [3]. Since access to data is also quite difficult in the case of AR accidents, it could also be worth trying to obtain data via social media.

## REFERENCES

- [1] Jem Bartholomew. 2022. Rising popularity of VR headsets sparks 31% rise in insurance claims. *The Guardian* (Feb. 2022). <https://www.theguardian.com/technology/2022/feb/12/rising-popularity-of-vr-headsets-sparks-31-rise-in-insurance-claims>
- [2] Emily Dao, Andreea Muresan, Kasper Hornbæk, and Jarrod Knibbe. 2021. Bad Breakdowns, Useful Seams, and Face Slapping: Analysis of VR Fails on YouTube.

- In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. ACM, Yokohama Japan, 1–14. <https://doi.org/10.1145/3411764.3445435>
- [3] Mara Faccio and John J. McConnell. 2020. Death by Pokémon GO: The Economic and Human Cost of Using Apps While Driving. *Journal of Risk and Insurance* 87, 3 (2020), 815–849. <https://doi.org/10.1111/jori.12301> \_eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/jori.12301>
  - [4] Sarah Faltaous, Joshua Neuwirth, Uwe Gruenefeld, and Stefan Schneeeggass. 2020. SaVR: Increasing Safety in Virtual Reality Environments via Electrical Muscle Stimulation. In *19th International Conference on Mobile and Ubiquitous Multimedia*. ACM, Essen Germany, 254–258. <https://doi.org/10.1145/3428361.3428389>
  - [5] Megan Joy Jones-Dellaportas, James Alexander Keitley, Richard Donald Bullough, and Blair Graham. 2022. Major traumatic injury sustained during use of a virtual reality (VR) headset. *Trauma* (2022), 14604086221100139.
  - [6] Bokyung Kye, Nara Han, Eunji Kim, Yeonjeong Park, and Soyoung Jo. 2021. Educational applications of metaverse: possibilities and limitations. *Journal of Educational Evaluation for Health Professions* 18 (Dec. 2021), 32. <https://doi.org/10.3352/jeehp.2021.18.32>
  - [7] Mark McGill, Daniel Boland, Roderick Murray-Smith, and Stephen Brewster. 2015. A dose of reality: Overcoming usability challenges in vr head-mounted displays. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*. 2143–2152.
  - [8] Clíodhna O'Connor and Helene Joffe. 2020. Intercoder reliability in qualitative research: debates and practical guidelines. *International journal of qualitative methods* 19 (2020), 1609406919899220.
  - [9] Joseph O'Hagan, Julie R Williamson, Mark McGill, and Mohamed Khamis. 2021. Safety, power imbalances, ethics and proxy sex: Surveying in-the-wild interactions between vr users and bystanders. In *2021 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. IEEE, 211–220.
  - [10] Kate Richards, Kai Wong, and Mansoor Khan. 2017. Augmented reality game related injuries. *New Horizons in Clinical Case Reports* 2 (2017), 27.
  - [11] Wen-Jie Tseng, Elise Bonnail, Mark McGill, Mohamed Khamis, Eric Lecolinet, Samuel Huron, and Jan Gugenheimer. 2022. The dark side of perceptual manipulations in virtual reality. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–15.
  - [12] Julius von Willich, Markus Funk, Florian Müller, Karola Marky, Jan Riemann, and Max Mühlhäuser. 2019. You Invaded my Tracking Space! Using Augmented Virtuality for Spotting Passersby in Room-Scale Virtual Reality. In *Proceedings of the 2019 on Designing Interactive Systems Conference*. ACM, San Diego CA USA, 487–496. <https://doi.org/10.1145/3322276.3322334>